M Zulfi Fahreza

Fullstack Software Engineer & AI Engineer

mrezalf0@gmail.com / reza@nexatriv.com
linkedin.com/in/m-zulfi-fahreza-552a1220b
github.com/v-reza
+62 896 5276 6610
Malang, East Java, Indonesia

Fullstack Software Engineer with 4+ years on production web systems for enterprise and government clients, currently leading a project team through an AI-assisted delivery workflow. Works across the stack: React and Next.js on the front, Spring Boot and Node.js behind. Designs RESTful and GraphQL APIs and the database schemas under them, and handles bug fixing, performance tuning and production troubleshooting. Also builds AI systems hands-on, including multi-agent orchestration, RAG pipelines and LLM fine-tuning for self-hosted agent tooling and an AI assessment engine.

WORK EXPERIENCE

Senior Software Engineer Jan 2026 - Present

MGG Software LTD., Singapore (remote); promoted from Fullstack Software Engineer

- Lead a 6-person project team across 3 concurrent client projects: own the plan and task allocation, run the technical decisions, and keep delivery on track across all three workstreams
- Own the client relationship at project level, running requirements and delivery discussions and committing the team to agreed scope and dates. 90% of milestones ship on the committed date
- Elicit and analyse requirements, then write the flows, UI and functional documentation the team and the client work from, across 20+ modules
- Carry ~40% of the team's delivery capacity as a hands-on contributor across the React and Vanilla JS frontend and the backend services
- Own product quality end to end: test planning, validation and code review across the team's work. 80% of releases are accepted by the client with no requirement-traceable defect
- Run the team's AI-assisted delivery workflow end to end, reviewing requirements, flows, UI, code, tests and documentation before release, and mentor 5 engineers on applying it. Cut feature rework by ~70%

Co-Founder & Software Engineer (Part-Time) Jan 2026 - Present

Nexatriv, Malang, Indonesia

- Co-founded a software and AI studio with two partners and own the engineering function across 3 product lines planned for 2026
- Architecting Nexalab, a self-hosted task management platform for deeply nested cross-team work, targeting 10,000+ tasks per workspace, with a free Solo tier instead of per-seat pricing
- Scoping an ERP suite for SME clients covering finance, inventory and HR, built as a modular monolith on PostgreSQL so each module ships on its own
- Set the studio's engineering standards: repo conventions, code review gates, and an AI-assisted delivery workflow reused across client and internal work

Fullstack Software Engineer Dec 2022 - Jan 2026

MGG Software LTD., Singapore (remote)

- Developed and maintained web applications in React and Spring Boot across 6+ platform modules
- Designed and implemented RESTful APIs for system integration, 25+ endpoints across 3 backend services
- Cut endpoint latency by ~40% on high-traffic modules through bug fixing, performance work and production troubleshooting
- Designed and maintained database schemas and relationships across multiple data sources
- Worked with BA, QA and DevOps teams to deliver features, and supported UAT and client feedback cycles
- Wrote technical and functional documentation for system features
- Provided technical consultation and proposed improvements or alternative solutions
- Translated business requirements into technical solutions and system designs

Fullstack Web Developer Apr 2022 - Sep 2022

ICUBE by SIRCLO, Yogyakarta, Indonesia

- Developed and maintained eCommerce applications in Magento (PHP) and Next.js progressive web apps for 4 enterprise retail clients
- Integrated GraphQL APIs into Next.js applications with Apollo Client, cutting round-trip API calls by ~40% on product listing pages
- Kept applications to security standards; passed penetration testing in development and staging with zero high-severity findings
- Optimized application performance and user experience, gaining hands-on experience in agile delivery

Backend Developer (Internship) May 2021 - Dec 2021

PT. Sekawan Media Informatika, Malang, Indonesia

- Built RESTful APIs from system design and workflow diagrams, 15+ endpoints across 3 client projects
- Handled bug fixing and debugging to keep application uptime above 99% across releases
- Supported penetration testing and automated testing with Cypress, raising regression coverage to ~70%
- Worked with the team to implement backend features to spec and improve API performance

PROJECTS

Assay - Active Socratic Learning Engine Next.js, Node.js, RAG, Google Cloud

- Built for AI Builder Cup 2026 (Google Cloud x Hack2skill, JAPAC) by a 3-developer squad in Track 4 (Adaptive Learning)
- Engineered a Socratic assessment engine: intent, role and an optional PDF produce a topic checklist and a chaptered syllabus, and each chapter stays locked until the user passes an interview quiz
- Implemented PDF-RAG grounding with optional Google Search grounding, plus a select-a-word interaction where the AI answers with a question instead of a summary
- Replaced a binary honeypot gate with a scored dimension (+2/-2/0/-1) after research showed hard gates punish honest learners; anti-cheat now runs as an on-device MediaPipe Focus Mode

AgentDeck - Self-Hosted AI Agent Fleet Orchestration Go, PostgreSQL 16, React 19, Docker

- Built a self-hosted orchestration board for AI agent fleets in Go and PostgreSQL 16. One binary, no sidecar services, deployed with Docker Compose
- Implemented a per-run and per-step cost ledger in integer micro-USD so no floating-point money crosses the boundary, keeping sub-cent spend at full precision instead of rounding it to $0.00
- Designed approval gates for sensitive agent actions with an inbox and audit trail, plus a four-role RBAC matrix with server-side tenant isolation (a foreign org returns 403, an unknown one 404)
- Shipped the React 19 + Vite SPA with Redux Toolkit, RTK Query and dnd-kit, covering identity, workspace and RBAC
- Wrote the test suite against a pinned PostgreSQL 16 image; migrations run on boot inside a single advisory-locked transaction

Nexalab - Deeply Nested Task Management Platform Rust, PostgreSQL, React

- Designing a self-hosted task management platform for deeply nested cross-team work: unlimited nesting depth, a dependency graph and role-scoped views
- Chose a Rust + PostgreSQL core for predictable latency, with an append-only activity log for auditable mutations
- Targeting a free, fully self-hostable Solo tier, with team roles as a feature rather than a paywall

Singapore Government System - Use of Parking Lots React, Spring Boot, Azure, Docker

- Developed a Singapore government system for allocating and using parking lots, covering lot inventory, allocation rules and usage reporting
- Built frontend and backend features in React and Spring Boot, integrating the API layer against existing government services
- Handled bug fixing, performance tuning and production troubleshooting under government security and compliance requirements
- Prepared technical and functional documentation and supported UAT with government stakeholders

Arche Framework - Internal AI Delivery Automation TypeScript, Node.js, Python, PostgreSQL, Docker, Multi-LLM

- Built an internal AI framework with 2 other engineers at MGG that turns a project's requirements into delivery artifacts: functional and technical documents, flowcharts and code scaffolding. Document turnaround went from days to hours
- Designed a provider-agnostic LLM layer over 4+ model providers, with per-task model routing, automatic failover on provider errors, and per-run token and cost accounting
- Engineered an agentic orchestration layer with a planner/executor split, tool calling, schema-constrained output and retry-with-repair, so every artifact comes out as validated structured output instead of free text
- Implemented a diagram pipeline that emits Mermaid and PlantUML flowcharts from the same requirement model the documents use, so diagrams and specs cannot drift apart
- Added code generation from the technical spec that scaffolds data models and API layers per project, gated behind human review before anything reaches the repository
- Wired the framework into the team's delivery workflow, with generated artifacts reviewed against requirements before client sign-off

Singapore Government System - Appeal Management System ReactJS, TypeScript, Vite, Spring Boot, Docker, Azure

- Contributed as a Fullstack Software Engineer to a Singapore government system managing appeal submission, review and decision workflows
- Owned development across the React frontend and Spring Boot backend
- Designed and implemented RESTful APIs for appeal processing, user roles and workflow management
- Translated business requirements into system design and data flow, and handled production troubleshooting to government compliance standards
- Guided developers through development and maintenance phases, and wrote technical documentation for onboarding

DatascripMall.ID - B2B/B2C eCommerce Platform PHP, Magento, Docker

- Contributed to a large-scale eCommerce platform serving B2B and B2C customers, with product catalog, shopping cart and checkout workflows
- Developed and maintained frontend and backend features, designed the APIs behind product management and transactions, and improved stability through bug fixing, optimization and production troubleshooting

Wacoal Philippines eCommerce Website PHP, Magento, Docker

- Maintained and enhanced an eCommerce platform for a global apparel brand, keeping the system stable and reliable in production
- Handled bug fixing, debugging and production troubleshooting, implemented enhancements from business and user requirements, and worked with BA, QA and DevOps teams on testing and deployment
- Monitored system health across development and production environments

Senheng Malaysia eCommerce Platform PHP, Magento, Next.js, Docker, PWA

- Maintained and enhanced a large-scale eCommerce platform for a leading consumer electronics retailer in Malaysia
- Handled bug fixing, debugging and production troubleshooting, implemented enhancements from business requirements and user feedback, and improved performance in high-traffic scenarios
- Worked with BA, QA and DevOps teams to analyse and resolve issues, and monitored system health across environments

Erhastore - ERHA Official Online Store PHP, Magento, Docker

- Maintained and enhanced the official eCommerce platform for a skincare brand
- Handled bug fixing, debugging and production troubleshooting to keep transaction flows stable, and implemented enhancements from business and user requirements
- Worked with BA, QA and DevOps teams to investigate and resolve issues across web platforms

SIPAS - e-Office Document Management System PHP Falcon, ReactJS, Docker, Jest

- Contributed as a Backend Developer to a web-based document management system handling incoming/outgoing letters, dispositions and digital archives
- Designed and developed RESTful APIs within a microservices architecture for document management, user roles and workflow processing
- Implemented letter tracking and disposition logic, optimized queries for document and transactional data, and handled bug fixing and backend troubleshooting
- Worked with frontend developers and BA/QA teams to keep integration and feature delivery smooth, and supported testing and deployment for stability and data integrity

CORE SKILLS

Programming Languages	TypeScript, JavaScript, Java, Python, PHP, Golang, Rust, Dart, SQL
Backend	Spring Boot, Node.js, NestJS, Laravel, Flask, PHP Falcon, GraphQL, Microservices, Hibernate
Frontend	ReactJS, Next.js, Vue.js, TailwindCSS, Shadcn/UI, Redux, Zustand, TanStack Query, Vite, PWA, Electron
Mobile	Flutter, React Native, Dart
Database	PostgreSQL, MySQL/MariaDB, MSSQL, Oracle, MongoDB, Redis, Firebase, Prisma ORM
DevOps	Docker, Kubernetes, CI/CD (GitHub Actions), Git, Nginx, Vercel, GCP, AWS, Azure, Linux/VPS, Sentry
Testing	Jest, Cypress, Playwright, Postman, Go Testing, Automated Testing, Test Planning, Code Review
AI Tools	Claude Code, Codex, OpenCode, GitHub Copilot, Hermes Agent, MCP, LLM Fine-tuning, RAG, Prompt Engineering, Multi-agent Orchestration, Multi-LLM Routing, Embeddings, Function Calling

LANGUAGES

- Bahasa Indonesia (Native)
- English (Professional Working)

EDUCATION

SMK Negeri 4 Malang Graduated: Sep 2022

Vocational High School Diploma in Software Engineering

CERTIFICATION / ACHIEVEMENT & AWARDS

- Participant, AI Builder Cup 2026 by Hack2Skill (Google Cloud), 2026
- Top 5 Winner (Programmer), Hackathon Semesta Scholarship 2023, PT. Sentra Vidya Utama, Jul 2023
- Participant, Special Bootcamp Fullstack Javascript by Sanbercode, Feb 2023
- React, HackerRank, Sep 2022
- Problem Solving, HackerRank, Sep 2022
- Mastering Flutter and Dart: Find House App, BuildWithAngga, Jul 2022
- Cisco Networking Academy, CPA: Programming Essentials in C++, Jan 2020